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1. Introduction to Discrete Mathematics and Syllabus

1.1 What is Discrete Mathematics?

Discrete Mathematics is the study of mathematical structures that are **discrete** (countable, separate, distinct) rather than continuous. It deals with objects such as integers, graphs, sets, relations and logical statements. By contrast, it excludes topics of “continuous mathematics” such as real numbers, calculus and Euclidean geometry.

Discrete Mathematics forms the mathematical foundation of computer science. It supplies the language and tools used to reason about algorithms, programs, data structures and networks.

1.2 Applications in Computer Science

- Advanced algorithms and data structures
- Programming language compilers and interpreters
- Computer networks, operating systems and computer architecture
- Database management systems
- Cryptography and error-correcting codes
- Graphics and animation algorithms, game engines
- Artificial Intelligence and automated theorem proving

In short, discrete mathematics underlies essentially the whole field of computing.

1.3 Syllabus of Unit I - Logic and Relations

Unit I (08 Hrs):

Logic: Propositions and logical operations, Conditional statements; Predicate Calculus — First order logic, universal and existential quantifiers; Proof Techniques — methods of proof, Mathematical induction.

Relations and Diagraphs: Properties of relations, composition of relations, closure operation on relations, equivalence relations and partitions, paths in relation and diagraphs, operations on relations, Transitive closure and Warshall's Algorithm. Partial Ordered Sets (POSET), Hasse diagram, external elements of partially ordered sets, Lattices.

The remaining units of the course (for reference) are:

- **Unit II - Combinatorics:** Combinations and permutations, enumerating combinations and permutations, Principle of Inclusion-Exclusion and Derangements, Generating functions, Recurrence relations and their solutions.
- **Unit III - Graph Theory:** What is a graph, matrices and isomorphism, special graphs, paths, cycles and trails, connected graphs, bipartite graphs, Eulerian paths and circuits, Hamiltonian paths and circuits, Fleury's algorithm, planar graphs, embedding and Euler's formula, colouring of graphs.

- **Unit IV - Trees and Networks:** Trees and basic properties, DFS and BFS algorithms, minimal spanning tree, shortest path problem, Kruskal's, Prim's and Dijkstra's algorithms, network flow problem, Ford-Fulkerson algorithm, Max-Flow-Min-Cut theorem.
- **Unit V - Algebraic Structures and Applications:** Binary operations, semi-groups and groups, subgroups, cosets, Lagrange's theorem, product and quotient semi-groups and groups, normal subgroup, homomorphism, coding of binary information and error detection, group codes, decoding and error correction.

Text Books (Reference)

- **T1.** Discrete Mathematical Structures — B. Kolman, R. Busby, S. C. Ross, Pearson / PHI, 6th Edition, 2009.
- **T2.** Discrete Mathematics for Computer Scientists and Mathematicians — J. L. Mott, A. Kandel, T. P. Baker, PHI, 2nd Edition, 2000.
- **T3.** Introduction to Graph Theory — Douglas B. West, PHI, 2nd Edition, 2003.

2. Logic and Propositions

2.1 Introduction to Mathematical Logic

Logic is the science of correct reasoning. It provides the foundation for mathematics, computer science, artificial intelligence, digital circuit design, database systems and many other disciplines. In mathematics, logic is used to determine whether an argument is valid and whether a statement is true or false. The word “logic” is derived from the Greek word *logos*, meaning reason or thought.

Logic helps us to:

- Analyse mathematical statements.
- Construct mathematical proofs.
- Verify computer programs.
- Design digital circuits.
- Develop algorithms.

Every mathematical theorem and proof is based on logical reasoning.

2.2 Propositions (Statements) and Non-Propositions

Definition - Proposition: A **proposition** (or statement) is a declarative sentence that has a definite truth value, either **True (T)** or **False (F)**, but not both simultaneously. In logic, only propositions can be analysed for truth or falsity.

Examples of propositions:

Proposition	Truth Value	Reason
7 is a prime number.	T	Only divisors of 7 are 1 and 7.
12 is an odd number.	F	12 is divisible by 2, hence even.
The Earth revolves around the Sun.	T	Astronomically true.
$2 + 3 = 8$	F	$2 + 3 = 5$, not 8.
Every square has four sides.	T	Definition of a square.

Non-Propositions: A sentence that does not have a definite truth value is **not** a proposition.

Sentence	Type	Why not a proposition
What is your name?	Question	Asks for information.
Close the door.	Command	Commands cannot be true or false.
What a beautiful day!	Exclamation	Expresses emotion, not a fact.
$x + 3 = 7$	Open sentence	Truth depends on the value of x.

For $x + 3 = 7$: when $x = 4$ the sentence is true, when $x = 5$ it is false — hence its truth value is not fixed.

2.3 Simple and Compound Propositions

Simple (Atomic) Proposition: A proposition that cannot be broken into smaller propositions.

Examples: “5 is prime”, “Today is Monday”, “Odisha is a state of India”.

Compound Proposition: A proposition formed by combining two or more propositions using logical connectives.

Examples: “It is raining and I carry an umbrella”, “Ram passed Mathematics or Physics”, “If it rains, then roads become wet”.

Truth values: every proposition has exactly one truth value — **T** (True) or **F** (False). For example: 5 is prime (T), 9 is prime (F), 10 is even (T), 11 is even (F).

2.4 Logical Connectives

Logical connectives are symbols used to combine propositions. Let

p : It is raining. q : I carry an umbrella.

Using logical connectives we can construct the compound propositions $p \wedge q$, $p \vee q$, $\neg p$, $p \rightarrow q$, $p \leftrightarrow q$.

The fundamental connectives are: (1) Negation, (2) Conjunction, (3) Disjunction, (4) Exclusive OR, (5) Implication, (6) Biconditional.

2.5 Negation, Conjunction, Disjunction and Exclusive OR

Negation (NOT)

Definition: The negation of a proposition p is denoted by $\neg p$ and means “not p ”.

Negation reverses the truth value.

Example: p : “Today is Monday”, $\neg p$: “Today is not Monday”. If p is true then $\neg p$ is false.

p	$\neg p$
T	F
F	T

Conjunction (AND)

Definition: The conjunction of p and q is $p \wedge q$, read “ p and q ”. A conjunction is **true only when both** propositions are true.

Example: p : “Ram passed Mathematics”, q : “Ram passed Physics”. Then $p \wedge q$ means “Ram passed Mathematics and Physics”. A scholarship requiring CGPA > 8.0 **AND** attendance $> 75\%$ is granted only when both conditions hold.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction (OR)

Definition: The disjunction of p and q is $p \vee q$, read “ p or q ”. In mathematics OR means **inclusive OR**.

Example: A student may submit the assignment online **OR** offline **OR** both.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Common Mistake: Students often think OR means exactly one. In mathematics, OR allows both statements to be true.

Exclusive OR (XOR)

Definition: Exclusive OR means **exactly one** proposition is true. Symbol: $p \oplus q$.

Example: A switch can be ON **or** OFF but not both.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

2.6 Implication and Biconditional

Implication (Conditional)

Definition: An implication has the form $p \rightarrow q$, read “If p , then q ”, where p is the **hypothesis** (antecedent, premise) and q is the **conclusion** (consequent).

Examples: “If a number is divisible by 4, then it is even”; “If a student studies regularly, then the student performs well”.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Important Observation: An implication is **false only when p is true and q is false**. This is the most important property of implication.

Biconditional

Definition: The biconditional is $p \leftrightarrow q$, read “ p if and only if q ” (p iff q). It means both $p \rightarrow q$ and $q \rightarrow p$ are true.

Example: An integer n is even if and only if $n = 2k$ for some integer k .

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

2.7 Tautology, Contradiction and Contingency

Tautology: A compound proposition that is **always true**. Example: $p \vee \neg p$ — the *Law of Excluded Middle*.

Contradiction: A compound proposition that is **always false**. Example: $p \wedge \neg p$ — the *Law of Contradiction*.

Contingency: A proposition that is **neither** always true nor always false. Example: $p \wedge q$; its truth depends on the values of p and q . Another example is $p \rightarrow \neg p$.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$	$p \rightarrow \neg p$
T	F	T	F	F
F	T	T	F	T

The first compound column is always T (tautology), the second always F (contradiction), the third varies (contingency).

2.8 Precedence of Logical Operators

When no parentheses are given, operators are applied in the order of precedence. Negation ($\neg$) binds most tightly, then $\wedge$, then $\vee$, then $\rightarrow$ and $\leftrightarrow$. Thus

$$p \vee q \rightarrow \neg r \quad \text{is equivalent to} \quad (p \vee q) \rightarrow \neg r$$

If the intended meaning is $p \vee (q \rightarrow \neg r)$, then parentheses must be used. Quantifiers have higher precedence than all logical operators (see Section 4).

Worked Example - Truth table for $p \vee q \rightarrow \neg r$:

p	q	r	$p \vee q$	$\neg r$	$(p \vee q) \rightarrow \neg r$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	F	T	T

A truth table with n propositional variables has 2^n rows.

3. Conditional Statements and Logical Equivalence

3.1 Conditional Statements (Implication)

A **conditional statement** has the form $p \rightarrow q$ and is read “If p , then q ”, where p is the **hypothesis** (antecedent, premise) and q is the **conclusion** (consequent, result).

Examples: (1) If a number is divisible by 4, then it is even. (2) If it rains, then the roads become wet. (3) If a student studies regularly, then the student performs well in examinations.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Understanding the Truth Table (all four cases)

- **Case 1 (T, T):** “If a number is divisible by 4, then it is even” — 8 is divisible by 4 and is even, so true.
- **Case 2 (T, F):** “If a number is divisible by 4, then it is odd” — 8 is divisible by 4 but not odd, so false. **This is the only situation in which an implication is false.**
- **Case 3 (F, T):** “If 5 is divisible by 4, then 5 is prime” — hypothesis is false, so the implication is considered true.
- **Case 4 (F, F):** “If 5 is divisible by 4, then 5 is even” — hypothesis is false, so the implication is considered true.

Examination Note: A conditional statement is **false only when the hypothesis is true and the conclusion is false.**

Different Ways of Expressing $p \rightarrow q$

- if p , then q | p implies q | if p , q
- p only if q | q unless $\neg p$ | q when p
- q whenever p | p is sufficient for q | q is necessary for p
- q follows from p | a necessary condition for p is q | a sufficient condition for q is p

3.2 Converse, Inverse and Contrapositive

From a conditional $p \rightarrow q$ we can form three new conditional statements:

Name	Form	How it is obtained
Conditional	$p \rightarrow q$	original

Converse	$q \rightarrow p$	interchange hypothesis and conclusion
Inverse	$\neg p \rightarrow \neg q$	negate both hypothesis and conclusion
Contrapositive	$\neg q \rightarrow \neg p$	interchange and negate both

Example: Original: “If a number is divisible by 4, then it is even” (True).

- **Converse:** “If a number is even, then it is divisible by 4” — **False**. Counterexample: 6 is even but not divisible by 4.
- **Inverse:** “If a number is not divisible by 4, then it is not even” — **False**. Counterexample: 6 is not divisible by 4 but is even.
- **Contrapositive:** “If a number is not even, then it is not divisible by 4” — **True**.

Important Theorem: A conditional statement and its **contrapositive are logically equivalent**:

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

Also, the **converse and inverse are logically equivalent** to each other.

3.3 Biconditional Statements

Definition: A biconditional statement $p \leftrightarrow q$ is read “p if and only if q” (p iff q). It means **both** $p \rightarrow q$ and $q \rightarrow p$ are true.

Example: “An integer n is even if and only if $n = 2k$ for some integer k.” This has two parts — (1) If n is even then $n = 2k$; (2) If $n = 2k$ then n is even. Both are true, so the biconditional is true.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Alternative ways to express “p iff q”: “p is necessary and sufficient for q”; “if p then q, and conversely”; “p iff q”.

3.4 Necessary and Sufficient Conditions

These concepts frequently appear in theorem statements.

Sufficient Condition: If $p \rightarrow q$, then **p is a sufficient condition for q**. Example: being divisible by 4 is sufficient for being even.

Necessary Condition: If $p \rightarrow q$, then **q is a necessary condition for p**. Example: being even is necessary for being divisible by 4.

Example: "If a student gets admission to a PhD program, then the student has a master's degree." A master's degree is **necessary**; admission to the PhD is **sufficient**.

3.5 Logical Equivalence

Definition: Two compound propositions are **logically equivalent** if they have identical truth values under every possible assignment of truth values. Notation: $p \equiv q$ (or $p \Leftrightarrow q$). Equivalently, $p \leftrightarrow q$ is a tautology.

Example - Show that $p \rightarrow q \equiv \neg p \vee q$:

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The columns for $p \rightarrow q$ and $\neg p \vee q$ are identical, hence $p \rightarrow q \equiv \neg p \vee q$.

3.6 Laws of Logical Equivalence

These laws are used to simplify logical expressions.

Law	Form
Identity	$p \wedge T \equiv p$ $p \vee F \equiv p$
Domination	$p \vee T \equiv T$ $p \wedge F \equiv F$
Idempotent	$p \vee p \equiv p$ $p \wedge p \equiv p$
Double Negation	$\neg(\neg p) \equiv p$
Negation	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$
Commutative	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
Associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
De Morgan's	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Additional equivalences involving conditionals: $p \rightarrow q \equiv \neg p \vee q \equiv \neg q \rightarrow \neg p$; $p \vee q \equiv \neg p \rightarrow q$; $p \wedge q \equiv \neg(p \rightarrow \neg q)$; $\neg(p \rightarrow q) \equiv p \wedge \neg q$; $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$; $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$.

3.7 De Morgan's Laws

De Morgan's Laws:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Example: “It is not true that Ram and Shyam both passed” is equivalent to “Ram failed or Shyam failed”.

Equivalence Proof 1: Show that $\neg(\neg p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.

$$\begin{aligned} & \neg(\neg p \vee (\neg p \wedge q)) \\ & \equiv \neg(\neg p) \wedge \neg(\neg p \wedge q) \quad \leftarrow \text{second De Morgan Law} \\ & \equiv p \wedge \neg(\neg p \wedge q) \quad \leftarrow \text{double negation} \\ & \equiv p \wedge (p \vee \neg q) \quad \leftarrow \text{De Morgan + double negation} \\ & \equiv p \wedge \neg q \quad \leftarrow \text{absorption} \end{aligned}$$

Equivalence Proof 2: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

$$\begin{aligned} & (p \wedge q) \rightarrow (p \vee q) \\ & \equiv \neg(p \wedge q) \vee (p \vee q) \quad \leftarrow p \rightarrow q \equiv \neg p \vee q \\ & \equiv (\neg p \vee \neg q) \vee (p \vee q) \quad \leftarrow \text{De Morgan} \\ & \equiv (\neg p \vee p) \vee (\neg q \vee q) \\ & \equiv T \vee T \equiv T \quad \leftarrow \text{tautology} \end{aligned}$$

4. Predicate Calculus (First-Order Logic)

4.1 Introduction to Predicate Logic

In propositional logic, statements are treated as indivisible units, e.g. “5 is a prime number”, “It is raining”, “Ram passed the examination”. Each statement is either true or false. However, many mathematical statements involve **variables** and cannot be expressed adequately using propositional logic, for example:

- Every student passed the examination.
- Some students own laptops.
- Every prime number greater than 2 is odd.
- There exists an integer whose square is 25.

Such statements require a more powerful logical system, called **Predicate Logic** or **First-Order Logic**.

Predicate Logic extends propositional logic by introducing: (1) Variables, (2) Predicates, (3) Quantifiers, (4) Functions, (5) Relations. It allows us to describe properties of objects and relationships among objects.

A predicate-logic statement may contain: **Variables** (x, y, z), **Constants** (1, 2, 5, a, b), **Predicates** ($P(x), Q(x,y)$), **Quantifiers** ($\forall, \exists$), and **Logical Connectives** ($\wedge, \vee, \neg, \rightarrow, \leftrightarrow$).

4.2 Predicates and Propositional Functions

Definition: A **predicate** is a statement involving one or more variables that becomes a proposition when values are assigned to those variables. It is also called a **propositional function**.

Examples:

- $P(x): x > 5$. Not a proposition until x is known. $P(8)$ is True, $P(2)$ is False.
- $Q(x): x$ is an even number. $Q(4) = \text{True}$, $Q(9) = \text{False}$.
- $R(x): x^2 = 25$. $R(5) = \text{True}$, $R(-5) = \text{True}$, $R(4) = \text{False}$.

Predicates with Two Variables

$P(x,y): x > y$. Then $P(5,2) = \text{True}$, $P(2,5) = \text{False}$, $P(3,3) = \text{False}$. Also $Q(x,y): x + y = 10$; $Q(6,4) = \text{True}$, $Q(3,8) = \text{False}$.

Predicates with Three Variables

$R(x,y,z): x + y = z$. Then $R(3,4,7) = \text{True}$, $R(2,-1,5) = \text{False}$, $R(5,5,10) = \text{True}$. Note that $R(x,y,z)$ itself is **not** a proposition (variables unspecified), but $R(3,4,7)$ **is** a proposition.

Example (from slides): Let “ $x + y = z$ ” be denoted by $R(x, y, z)$ with the domain the integers. Then $R(2, -1, 5)$ is F, $R(3, 4, 7)$ is T, and $R(x, 3, z)$ is *not* a proposition. Similarly for $Q(x, y, z): x - y = z$, we get $Q(2, -1, 3) = T$, $Q(3, 4, 7) = F$.

4.3 Domain of Discourse and Open Sentences

Domain of Discourse: The set from which variables take their values is called the **domain of discourse** or **universe of discourse** (often denoted U).

Common domains: Natural numbers $N = \{1, 2, 3, 4, \dots\}$; Integers $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$; Real numbers $\mathbb{R}$; positive integers $\{1, 2, 3, \dots\}$; students in a class; employees in a company; vertices of a graph.

Importance of Domain: The truth value of a predicate often depends on the domain. Example: $P(x): x^2 = 4$. Over the integers the solutions are $x = 2$ and $x = -2$; over the natural numbers only $x = 2$; over the reals $x = \pm 2$. Thus changing the domain may change the meaning and truth of statements.

Open Sentence: A statement containing variables whose truth value is not yet determined. Example: $x + 3 = 7$ (true when $x = 4$, false when $x = 2$); $x > y$ (truth depends on x and y).

4.4 Well-Formed Formulas (WFF)

Definition: A **well-formed formula** is a logically meaningful expression constructed according to the rules of logic.

Well-formed: $P(x)$, $P(x) \wedge Q(x)$, $P(x) \rightarrow Q(x)$, $\forall x P(x)$, $\exists x Q(x)$.

Not well-formed: $\wedge P(x)$, $\rightarrow \rightarrow P(x)$ — these violate the rules of logical syntax.

4.5 Universal and Existential Quantifiers

Predicates become complete mathematical statements through the use of **quantifiers**, which tell us how many elements satisfy a predicate. The two fundamental quantifiers are the **Universal Quantifier** and the **Existential Quantifier**.

Universal Quantifier ($\forall$): Read as “For all” or “For every”. Form: $\forall x P(x)$ — $P(x)$ is true for **every** element in the domain.

Examples:

- For every real number x , $x^2 \geq 0$ — **True** (a square is never negative).
- For every integer x , x is even — **False** (counterexample $x = 3$).
- For every positive integer n , $n + 1 > n$ — **True**.
- If $P(x)$ is " $x > 0$ " and U is the positive integers, then $\forall x P(x)$ is true; if U is the integers, it is false.

Existential Quantifier ($\exists$): Read as "There exists" or "There is at least one". Form: $\exists x P(x)$ — at least one element satisfies $P(x)$.

Examples:

- There exists an integer x such that $x^2 = 25$ — **True** ($x = 5$ or $x = -5$).
- There exists a prime number that is even — **True** (the number 2).
- There exists a natural number less than zero — **False**.
- If $P(x)$ is " x is even" and U is the integers, then $\exists x P(x)$ is true.

Universal vs Existential: "Every student passed" requires all students to satisfy the condition; "Some student passed" requires only one. The truth value of $\forall x P(x)$ and $\exists x P(x)$ depends on **both** the propositional function $P(x)$ and the domain U .

4.6 Counterexamples

Key Idea: To show that a **universal** statement is false, it is sufficient to find **one counterexample**.

Example: "Every prime number is odd" is false, since 2 is prime and even. Likewise, "Every student scored above 90%" is disproved by a single student scoring below 90%.

4.7 Translating English into Predicate Logic

Translation is one of the most important examination topics. First decide the domain U , then define propositional functions.

Examples:

- "Every student in this class has taken a course in Java." Let $J(x)$: x has taken Java, U = students in this class $\rightarrow \forall x J(x)$.
- "Some student in this class has taken a course in Java." $\rightarrow \exists x J(x)$.
- "Every prime number greater than 2 is odd." Let $P(x)$: x is prime, $O(x)$: x is odd $\rightarrow \forall x ((P(x) \wedge x > 2) \rightarrow O(x))$.
- "There exists an even prime number." $\rightarrow \exists x (x \text{ is prime} \wedge x \text{ is even})$.

4.8 Negation of Quantified Statements

Negating quantified statements is extremely important in proofs.

De Morgan's Laws for Quantifiers:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Examples:

- Statement: "All students passed." → Negation: "At least one student failed."
- Statement: "Every integer is positive." → Negation: "There exists an integer that is not positive."
- Statement: "There exists a student who scored 100%." → Negation: "No student scored 100%."
- Statement: "There exists an even prime greater than 2." → Negation: "Every prime greater than 2 is not even."

4.9 Nested Quantifiers and Order of Quantifiers

Nested Quantifiers: A statement containing more than one quantifier is a nested quantified statement. Example: "Every real number has an inverse" is $\forall x \exists y (x + y = 0)$, with domain the reals.

Order of Quantifiers: The order is extremely important. In general, "for every x there exists y" is **NOT** equivalent to "there exists y for every x".

Examples:

- "For every real number x, there exists a real number y such that $x + y = 0$ " — **True** (choose $y = -x$).
- "There exists a real number y such that for every real number x, $x + y = 0$ " — **False** (no single y works for all x).
- "Every student has a favourite teacher" ($\forall$ student $\exists$ teacher) may be true, whereas "There exists a teacher who is everyone's favourite" ($\exists$ teacher $\forall$ student) is a completely different — and possibly false — statement.

Practice (from slides): Let U be the real numbers and $P(x,y): x \cdot y = 0$. Then: $\forall x \forall y P(x,y) \rightarrow$ **False**; $\forall x \exists y P(x,y) \rightarrow$ **True**; $\exists x \forall y P(x,y) \rightarrow$ **True**; $\exists x \exists y P(x,y) \rightarrow$ **True**.

Precedence of quantifiers: quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators. So $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$, which is different from $\forall x (P(x) \vee Q(x))$.

Applications of predicate logic: mathematics, computer programming, artificial intelligence, database systems, software verification, graph theory, automata theory, formal language theory.

5. Rules of Inference and Valid Arguments

5.1 Arguments, Valid and Invalid Arguments

In mathematics we frequently derive conclusions from known facts. The process of drawing conclusions from premises according to accepted logical rules is called **inference**, and the rules used are called **Rules of Inference**. They form the foundation of mathematical proofs, computer programming, artificial intelligence, automated theorem proving and database query systems.

Argument: An argument consists of (1) one or more **premises** and (2) a **conclusion**.

Premise 1, Premise 2, ..., Premise n $\therefore$ Conclusion

An argument is **valid** if the premises imply the conclusion, i.e. the conclusion necessarily follows from the premises. If the premises are $p_1, p_2, \dots, p_n$ and the conclusion is q , then the argument is valid iff $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.

Valid argument: Premise 1: All humans are mortal. Premise 2: Socrates is a human.
Conclusion: Socrates is mortal. The conclusion must be true whenever the premises are true, so the argument is valid.

Invalid argument: Premise 1: If it rains, then the roads become wet. Premise 2: The roads are wet. Conclusion: It is raining. This is invalid because roads may become wet for other reasons.

5.2 Rules of Inference (Propositional Logic)

(1) Modus Ponens

$$\begin{array}{c} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Corresponding tautology: $(p \wedge (p \rightarrow q)) \rightarrow q$. If p is true and p implies q , then q must be true.

Premise 1: A student studies regularly. Premise 2: If a student studies regularly, then the student passes. $\therefore$ The student passes.

Also: 10 is divisible by 5; if a number is divisible by 5 then it is divisible by 1; $\therefore$ 10 is divisible by 1.

(2) Modus Tollens

$$\begin{array}{c} p \rightarrow q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Corresponding tautology: $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$. If p implies q and q is false, then p must also be false.

Premise 1: If it is snowing, then I will study Discrete Mathematics. Premise 2: I am not studying Discrete Mathematics. $\therefore$ It is not snowing.

Also: If a number is divisible by 6 then it is divisible by 3; the number is not divisible by 3; $\therefore$ the number is not divisible by 6.

(3) Hypothetical Syllogism

$$\begin{array}{c} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

If I study Mathematics, I will pass the examination. If I pass the examination, I will get a job. $\therefore$ If I study Mathematics, I will get a job.

Mathematical: If n is divisible by 8 then n is divisible by 4; if n is divisible by 4 then n is divisible by 2; $\therefore$ if n is divisible by 8 then n is divisible by 2.

(4) Disjunctive Syllogism

$$\begin{array}{c} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

Corresponding tautology: $(\neg p \wedge (p \vee q)) \rightarrow q$.

I will study Mathematics or Physics. I will not study Mathematics. $\therefore$ I will study Physics.

A number is even or odd; the number is not even; $\therefore$ the number is odd.

(5) Addition

$$\begin{array}{c} p \\ \hline \therefore p \vee q \end{array}$$

Corresponding tautology: $p \rightarrow (p \vee q)$. If p is true then $p \vee q$ must be true regardless of q .

I will study Mathematics. $\therefore$ I will study Mathematics or visit Puri.

(6) Simplification

$$\begin{array}{c} p \wedge q \\ \hline \therefore p \end{array}$$

Corresponding tautology: $(p \wedge q) \rightarrow p$.

I study Mathematics and Physics. $\therefore$ I study Mathematics. Also: 10 is even and positive $\rightarrow$ 10 is even.

(7) Conjunction

$$\begin{array}{c} p \\ q \\ \hline \therefore p \wedge q \end{array}$$

Corresponding tautology: $((p) \wedge (q)) \rightarrow (p \wedge q)$.

I study Mathematics. I study Physics. $\therefore$ I study Mathematics and Physics.

(8) Resolution

$$\begin{array}{c} \neg p \vee r \\ p \vee q \\ \hline \therefore q \vee r \end{array}$$

Corresponding tautology: $((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$.

Either I do not study Mathematics or I study English. Either I study Mathematics or I study Databases. $\therefore$ I study English or Databases.

5.3 Common Logical Fallacies

A **fallacy** is an incorrect form of reasoning.

Fallacy 1 - Affirming the Consequent:

$$\begin{array}{c} p \rightarrow q \\ q \\ \hline \therefore p \text{ (INVALID)} \end{array}$$

Example: If it rains, roads become wet. Roads are wet. $\therefore$ It is raining — incorrect, because roads may be wet due to cleaning or a water leak.

Fallacy 2 - Denying the Antecedent:

$$\begin{array}{c} p \rightarrow q \\ \neg p \\ \hline \therefore \neg q \text{ (INVALID)} \end{array}$$

Example: If it rains, roads become wet. It is not raining. $\therefore$ Roads are not wet — incorrect, because roads may still be wet for another reason.

5.4 Quantified Rules of Inference (UI, UG, EI, EG)

When quantifiers are involved, additional inference rules are required.

Universal Instantiation (UI): If a statement is true for all elements, then it is true for any particular element.

$$\frac{\forall x P(x)}{\therefore P(a)}$$

Example: All humans are mortal; Socrates is a human; $\therefore$ Socrates is mortal.

Mathematical: $\forall x (x + 0 = x)$; $\therefore 5 + 0 = 5$.

Universal Generalization (UG): If $P(x)$ is proved for an **arbitrary** element x , then $P(x)$ is true for all elements. Example: let x be any real number and prove $x + 0 = x$; $\therefore \forall x (x + 0 = x)$.

Existential Instantiation (EI): If there exists an element satisfying a property, we may choose one such element.

$$\frac{\exists x P(x)}{\therefore P(a) \text{ for some particular } a}$$

Example: There exists a student who scored 100%; let that student be A; then A scored 100%.

Existential Generalization (EG): If a specific element satisfies a property, then at least one such element exists. Example: 5 is prime; $\therefore$ there exists a prime number. Also: 16 is a perfect square; $\therefore$ there exists a perfect square.

Valid argument with quantifiers: Premise 1: Every student in this class has passed Mathematics. Premise 2: Rahul is a student in this class. $\therefore$ Rahul has passed Mathematics (uses Universal Instantiation).

Another: Every prime number greater than 2 is odd; 11 is prime and greater than 2; $\therefore$ 11 is odd.

5.5 Constructing Formal Proofs

To construct a proof: **Step 1** list all premises; **Step 2** identify applicable inference rules; **Step 3** apply the rules systematically; **Step 4** reach the required conclusion.

Example: Premises: (1) If it rains, then roads become wet. (2) If roads become wet, traffic slows down. (3) It rains. Conclusion: Traffic slows down.

From (1) and (3) $\rightarrow$ Roads become wet. (Modus Ponens)

From "roads become wet" and (2) $\rightarrow$ Traffic slows down. (Modus Ponens)

Hence proved.

Classic example (home by sunset): Hypotheses:

- It is not sunny this afternoon and it is colder than yesterday.
- We will go swimming only if it is sunny.
- If we do not go swimming, then we will take a canoe trip.
- If we take a canoe trip, then we will be home by sunset.

Conclusion: "We will be home by sunset." Choose p: sunny; q: colder than yesterday; r: go swimming; s: take a canoe trip; t: home by sunset. Translation and proof:

$\neg p \wedge q$ Premise $\rightarrow \neg p$ (Simplification)
 $r \rightarrow p$ Premise, with $\neg p \rightarrow \neg r$ (Modus Tollens)
 $\neg r \rightarrow s$ Premise, with $\neg r \rightarrow s$ (Modus Ponens)
 $s \rightarrow t$ Premise, with $s \rightarrow t$ (Modus Ponens)
 $\therefore t$ – we will be home by sunset.

Applications of rules of inference: mathematical proofs, computer programming, logic circuit design, artificial intelligence, automated theorem proving, database systems, formal verification of software, cryptography.

6. Proof Techniques and Mathematical Induction

6.1 Introduction to Mathematical Proofs

Mathematics is based on logical reasoning. A mathematical statement is accepted as true only after a valid proof has been provided. A **proof** is a sequence of logically justified statements that establishes the truth of a mathematical proposition. It begins with known facts, definitions, axioms or previously established theorems and proceeds through logical reasoning to reach the desired conclusion.

Why proofs are important: they verify mathematical statements, develop logical reasoning skills, establish general results, explain *why* a statement is true, and help avoid incorrect conclusions.

Motivating example: Statement — “The sum of two even integers is even.” Checking a few examples ($2 + 4 = 6$, $8 + 10 = 18$) is **not sufficient**. A proof must establish the result for *all* even integers.

The most common proof techniques are: (1) Direct Proof, (2) Proof by Contrapositive, (3) Proof by Contradiction, (4) Proof by Cases, (5) Biconditional Proof, (6) Existence Proof, (7) Uniqueness Proof, (8) Mathematical Induction, (9) Strong Mathematical Induction.

6.2 Direct Proof

Definition: In a direct proof, we assume the hypothesis is true and use definitions, axioms and logical reasoning to derive the conclusion. Structure: assume $p \rightarrow$ use reasoning $\rightarrow$ show q .

Theorem 1: The sum of two even integers is even.

Let $a = 2m$ and $b = 2n$ where m, n are integers.

Then $a + b = 2m + 2n = 2(m + n)$.

Since $m + n$ is an integer, $a + b$ is divisible by 2. Hence $a + b$ is even. $\therefore$ proved.

Theorem 2: The sum of two odd integers is even.

Let $a = 2m + 1$, $b = 2n + 1$.

$a + b = (2m + 1) + (2n + 1) = 2m + 2n + 2 = 2(m + n + 1)$.

Since $m + n + 1$ is an integer, $a + b$ is even. $\therefore$ proved.

Theorem 3: If n is divisible by 6, then n is divisible by 3.

Suppose $n = 6k$ for some integer k . Then $n = 3(2k)$. Since $2k$ is an integer, n is divisible by 3. $\therefore$ proved.

Theorem 4: If n is an odd integer, then n^2 is odd.

Assume n is odd. Then $n = 2k + 1$ for an integer k .

$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1$, where $r = 2k^2 + 2k$ is an integer.

Hence n^2 is odd. $\therefore$ proved.

Theorem 5: The sum of two rational numbers is rational.

(Definition: r is rational if there exist integers p, q with $q \neq 0$ such that $r = p/q$.)

Assume r and s are rational: $r = p/q$, $s = t/u$ with $q \neq 0$, $u \neq 0$.

$r + s = p/q + t/u = (pu + qt)/(qu) = v/w$, where $v = pu + qt$ and $w = qu \neq 0$ are integers.

Hence $r + s$ is rational. $\therefore$ proved.

6.3 Proof by Contrapositive

Definition: To prove “If p , then q ” we instead prove “If not q , then not p ”. This is valid because a statement and its contrapositive are logically equivalent. (Also called an **indirect proof**.)

Theorem 1: If n^2 is odd, then n is odd.

We prove the contrapositive: assume n is even, so $n = 2k$.

Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is even.

Hence if n is even, n^2 is even. Therefore if n^2 is odd, n must be odd. $\therefore$ proved.

Theorem 2: If $3n + 2$ is odd, then n is odd.

Assume n is even, so $n = 2k$. Substituting:

$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$, which is even.

Hence if $3n + 2$ is odd, n must be odd. $\therefore$ proved.

6.4 Proof by Contradiction

Definition: We assume the statement to be false and derive an impossibility. Since the assumption leads to a contradiction, it must be false; therefore the original statement is true. Structure: assume the theorem is false $\rightarrow$ derive a contradiction $\rightarrow$ conclude the theorem is true.

Theorem 1: $\sqrt{2}$ is irrational.

Assume $\sqrt{2}$ is rational. Then $\sqrt{2} = a/b$ where a, b are integers with no common factor ($b \neq 0$).

Squaring: $2 = a^2/b^2 \Rightarrow a^2 = 2b^2$. Thus a^2 is even, hence a is even. Let $a = 2k$.

Substituting: $4k^2 = 2b^2 \Rightarrow b^2 = 2k^2$. Hence b is also even.

So both a and b are even – contradicting that they have no common factor.

Therefore $\sqrt{2}$ is irrational. $\therefore$ proved.

Theorem 2: There are infinitely many prime numbers.

Assume there are only finitely many primes $p_1, p_2, \dots, p_n$.

Consider $N = p_1 p_2 \dots p_n + 1$.

When N is divided by any listed prime, the remainder is 1. So none of the listed primes divides N .

Hence N has a new prime factor – a contradiction.

Therefore infinitely many primes exist. $\therefore$ proved.

6.5 Proof by Cases

Definition: Sometimes a statement naturally divides into several cases. If the statement is true in every case, it is true in general.

Theorem 1: For every integer n , $n(n + 1)$ is even.

Case 1: n is even $\rightarrow n(n + 1)$ is even.

Case 2: n is odd $\rightarrow n + 1$ is even $\rightarrow n(n + 1)$ is even.

True in all cases. $\therefore$ proved.

Theorem 2: $|n| \geq 0$ for every integer n .

Case 1: $n \geq 0 \rightarrow |n| = n \geq 0$.

Case 2: $n < 0 \rightarrow |n| = -n > 0$.

Therefore $|n| \geq 0$. $\therefore$ proved.

6.6 Biconditional, Existence and Uniqueness Proofs

Biconditional Proof: To prove $p \leftrightarrow q$, we prove both $p \rightarrow q$ and $q \rightarrow p$.

Theorem: An integer n is odd if and only if n^2 is odd.

Part 1 (n odd $\rightarrow n^2$ odd): $n = 2k + 1 \Rightarrow n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd.

Part 2 (n^2 odd $\rightarrow n$ odd): by contrapositive, if n were even then n^2 would be even – contradiction. Hence n is odd.

Therefore n is odd iff n^2 is odd. $\therefore$ proved.

Existence Proof: Demonstrates that at least one object satisfying a property exists.

Example: "There exists an even prime number" — 2 is prime and even, so such a number exists. Example: "There exists a real number x such that $x^2 = 25$ " — $x = 5$ satisfies it.

Uniqueness Proof: Shows that **exactly one** object satisfies a property. Example: the equation $x + 3 = 5$ has a unique solution. Proof: $x = 2$ satisfies it; suppose y is another solution, then $y + 3 = 5 \Rightarrow y = 2$, so $y = x$. Hence the solution is unique.

6.7 Common Errors in Proofs

A proof is valid only if every step is logically justified.

False proof that $1 = 2$:

Assume $a = b$.

Then $a^2 = ab$.

Subtract b^2 : $a^2 - b^2 = ab - b^2$.

Factor: $(a - b)(a + b) = b(a - b)$.

Divide by $(a - b)$: $a + b = b$.

Since $a = b$, $2b = b$. Therefore $2 = 1$.

Error: since $a = b$ we have $a - b = 0$, so the proof divides by zero, which is undefined. Hence the argument is invalid.

6.8 Mathematical Induction

Mathematical induction is a proof technique used to establish statements involving positive integers. It is based on the principle: "If the first domino falls and each domino knocks down the next one, then all dominoes fall."

Principle of Mathematical Induction: To prove $P(n)$ for all $n \geq 1$:

1. **Basis Step:** Show $P(1)$ is true.
2. **Induction Hypothesis:** Assume $P(k)$ is true.
3. **Inductive Step:** Show $P(k + 1)$ is true.

Then $P(n)$ is true for all positive integers n .

Example 1: Prove $1 + 2 + 3 + \dots + n = n(n + 1)/2$.

Basis: $n = 1$: LHS = 1, RHS = $1 \cdot 2/2 = 1$. True.

Hypothesis: assume $1 + 2 + \dots + k = k(k + 1)/2$.

Inductive step:

$$1 + 2 + \dots + k + (k + 1) = k(k + 1)/2 + (k + 1)$$

$$= (k + 1)(k + 2)/2 \text{ — which is the formula for } k + 1.$$

Hence true for $k + 1$. $\therefore$ proved for all n .

Example 2: Prove $n < 2^n$ for all positive integers n .

Basis: $1 < 2$. True.

Hypothesis: assume $k < 2^k$.

Inductive step: $k + 1 \leq 2k < 2 \cdot 2^k = 2^{k+1}$. Hence true.

Example 3: Prove $n^3 - n$ is divisible by 3.

Basis: $1^3 - 1 = 0$, divisible by 3.

Inductive step: $(k + 1)^3 - (k + 1) = (k^3 - k) + 3(k^2 + k)$.

Both terms are divisible by 3, so the sum is divisible by 3. $\therefore$ proved.

Example 4 (slides): Prove $2^n < n!$ for every integer $n \geq 4$.

Basis: $P(4)$: $2^4 = 16 < 4! = 24$. True. (Note $P(0)$, $P(1)$, $P(2)$, $P(3)$ are all false, so the basis is $P(4)$.)

Inductive step: $2^{k+1} = 2 \cdot 2^k < 2 \cdot k! < (k + 1) \cdot k! = (k + 1)!$. $\therefore$ proved.

6.9 Strong Mathematical Induction

Definition: In ordinary induction we assume only $P(k)$. In **strong induction** we assume $P(1)$, $P(2)$, ..., $P(k)$ are **all** true, and then prove $P(k + 1)$.

Example: Every integer greater than 1 can be expressed as a product of prime numbers.

Basis: 2 is prime.

Hypothesis: assume every integer from 2 to k can be written as a product of primes.

Inductive step: consider $k + 1$. If prime, done. If composite, $k + 1 = ab$ with $a, b < k + 1$.

By the induction hypothesis, a and b are products of primes. Therefore $k + 1$ is also a product of primes. $\therefore$ proved.

These proof techniques form the foundation of higher mathematics, graph theory, abstract algebra, automata theory, number theory and theoretical computer science.

7. Relations and Their Properties

7.1 Binary Relations, Domain, Range and Inverse

Definition: Let A and B be two arbitrary sets. A **binary relation** from A to B is a subset **f** of $A \times B$. If $A = B$, f is called a relation **on** A. If $(x, y) \in f$ we write $x f y$ (or $x R y$) and say “x is related to y through f”.

Example: Let $A = \{0,1,2\}$ and $B = \{a,b\}$. Then $\{(0,a), (0,b), (1,a), (2,b)\}$ is a relation from A to B.

Note: Since $\emptyset \subseteq A \times B$, the empty set $\emptyset$ is a relation from A to B; also, since $A \times B \subseteq A \times B$, the whole set $A \times B$ is a relation from A to B.

Domain, Range and Inverse:

- **Domain of f** = $\{x \in A : (x, y) \in f \text{ for some } y \in B\}$.
- **Range of f** = $\{y \in B : (x, y) \in f \text{ for some } x \in A\}$.
- **Inverse of f** = $f^{-1} = \{(y, x) : (x, y) \in f\}$.

Example: Let $A = B = \{1,2,3,4,5\}$ and (i) $f = \{(x,y) \in A \times B : x = y\}$; (ii) $f = \{(x,y) \in A \times B : x < y\}$. Find f, its domain and range.

7.2 One-one, One-many and Many-one Relations

- **One-one:** if $(a, b) \in f$ and $(a, b') \in f$ then $b = b'$ (each first element maps to a single, distinct second element).
- **One-many:** if $(a, b) \in f$ and $(a, b') \in f$ then $b \neq b'$ (one first element maps to more than one second element).
- **Many-one:** if $(a, b) \in f$ and $(a', b) \in f$ then $a \neq a'$ (several first elements map to the same second element).

7.3 Properties: Reflexive, Symmetric, Transitive, Antisymmetric

Reflexive: A relation f on A is **reflexive** if $a f a$, i.e. $(a, a) \in f$ for every $a \in A$.

Symmetric: f is **symmetric** if $(a, b) \in f \Rightarrow (b, a) \in f$, equivalently $f^{-1} = f$.

Transitive: f is **transitive** if $(a, b) \in f$ and $(b, c) \in f \Rightarrow (a, c) \in f$.

Antisymmetric: f is **antisymmetric** if $(a, b) \in f$ and $(b, a) \in f$ together imply $a = b$.

Examples from everyday life:

- Parallelism of two lines is **reflexive**; perpendicularity of two lines is **symmetric**; parallelism of three lines is **transitive**.
- The **fatherhood** relation is neither reflexive nor symmetric nor transitive.
- The **ancestor** relation satisfies only the transitive condition.
- The **spouse** relation satisfies only the symmetric condition.
- The **subset** relation of two sets satisfies the antisymmetric property; it is a partial ordering relation.
- The **division** relation is not an equivalence relation.

7.4 Equivalence, Partial Ordering and Total Ordering

Equivalence Relation: f is an equivalence relation if it is (i) reflexive, (ii) symmetric, (iii) transitive.

Partial Ordering Relation: f is a partial ordering relation if it is (i) reflexive, (ii) antisymmetric, (iii) transitive.

Total Ordering: f on A is a total ordering if it is a partial ordering and, for all $a, b \in A$, either $(a, b) \in f$ or $(b, a) \in f$.

7.5 Combining Relations

Given two relations R_1 and R_2 , we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, $R_2 - R_1$, and the symmetric difference $R_1 \oplus R_2 = (R_1 \cup R_2) - (R_1 \cap R_2)$.

Example: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$. With $R_1 = \{(1,1),(2,2),(3,3)\}$ and $R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$:

$$R_1 \cup R_2 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$$

$$R_1 \cap R_2 = \{(1,1)\}$$

$$R_1 - R_2 = \{(2,2), (3,3)\}$$

Example: Let R_1 be the “less than” relation and R_2 the “greater than” relation on the reals: $R_1 = \{(x,y) : x < y\}$, $R_2 = \{(x,y) : x > y\}$. Then $R_1 \cup R_2$ is “ $\neq$ ”, $R_1 \cap R_2 = \emptyset$, $R_1 - R_2 = R_1$, $R_2 - R_1 = R_2$, and $R_1 \oplus R_2 = “\neq”$.

7.6 Composition and Powers of a Relation

Composition: Suppose R_1 is a relation from A to B and R_2 is a relation from B to C. Then the **composition** of R_2 with R_1 , written $R_2 \circ R_1$, is a relation from A to C where if $(a,b) \in R_1$ and $(b,c) \in R_2$ then $(a,c) \in R_2 \circ R_1$.

$$A \xrightarrow{R_1} B \xrightarrow{R_2} C \Rightarrow A \xrightarrow{R_2 \circ R_1} C$$

Example: $R_1 = \{(1,1),(1,4),(2,3),(3,1),(3,4)\}$ and $R_2 = \{(1,0),(2,0),(3,1),(3,2),(4,1)\}$. Then $R_2 \circ R_1 = \{(1,0),(1,1),(2,1),(2,2),(3,0),(3,1)\}$.

Powers of a Relation: Let R be a binary relation on A. The powers R^n are defined recursively: Basis step $R^1 = R$; Recursive step $R^{n+1} = R^n \circ R$.

7.7 Representing Relations - Matrices

A relation between finite sets can be represented using a **zero-one matrix**. The relation R is represented by the matrix $M_R = [m_{ij}]$, where the (i,j) entry is 1 when a_i is related to b_j , and 0 otherwise.

Example 1: Let $A = \{1,2,3\}$ and $B = \{1,2\}$. Let R be the relation from A to B with $(a,b) \in R$ if $a > b$. Since $R = \{(2,1),(3,1),(3,2)\}$, the matrix is:

	1	2
1	0	0
2	1	0
3	1	1

Example 2: Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, \dots, b_5\}$. Given the matrix, the ordered pairs are those (a_i, b_j) with $m_{ij} = 1$, e.g. $R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}$.

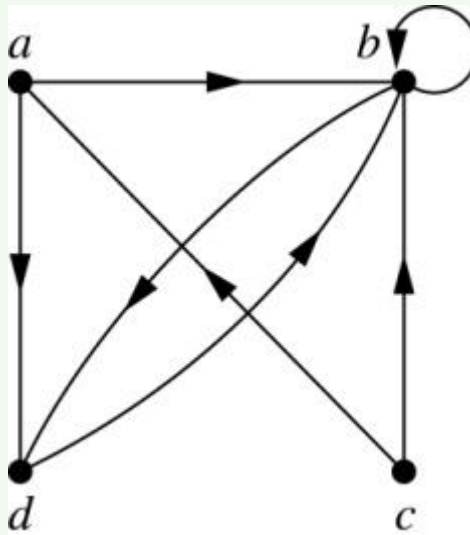
Matrices of Relations on a Set:

- If R is **reflexive**, all elements on the main diagonal of M_R are 1.
- R is **symmetric** iff $m_{ij} = 1$ whenever $m_{ji} = 1$ (the matrix is symmetric).
- R is **antisymmetric** iff $m_{ij} = 0$ or $m_{ji} = 0$ whenever $i \neq j$.

7.8 Representing Relations - Digraphs

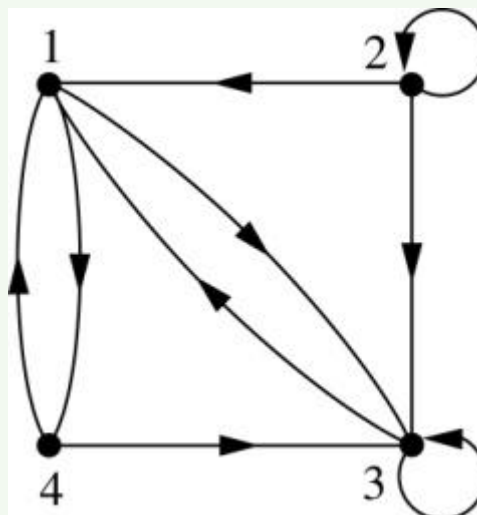
Definition: A **directed graph**, or **digraph**, consists of a set V of vertices (nodes) together with a set E of ordered pairs of elements of V called edges (arcs). The vertex a is the *initial vertex* of the edge (a,b) , and b is the *terminal vertex*. An edge of the form (a,a) is called a **loop**.

Example: A drawing of the directed graph with vertices a, b, c, d and edges $(a,b), (a,d), (b,b), (b,d), (c,a), (c,b), (d,b)$:



A digraph with vertices a, b, c, d and a loop at b .

Example: What are the ordered pairs in the relation represented by this directed graph?

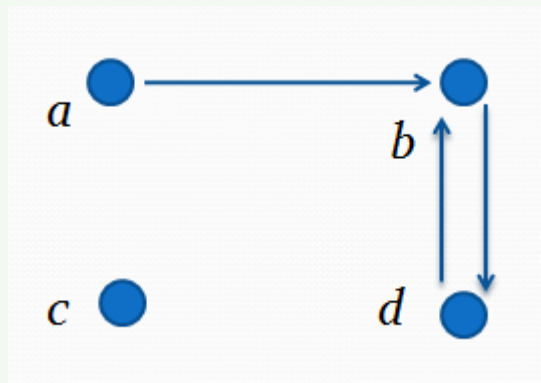


A digraph on vertices $1, 2, 3, 4$.

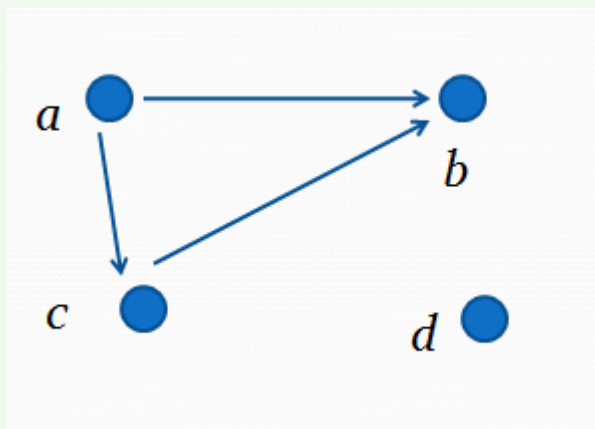
Solution: $(1,3), (1,4), (2,1), (2,2), (2,3), (3,1), (3,3), (4,1), (4,3)$.

Determining Properties from a Digraph:

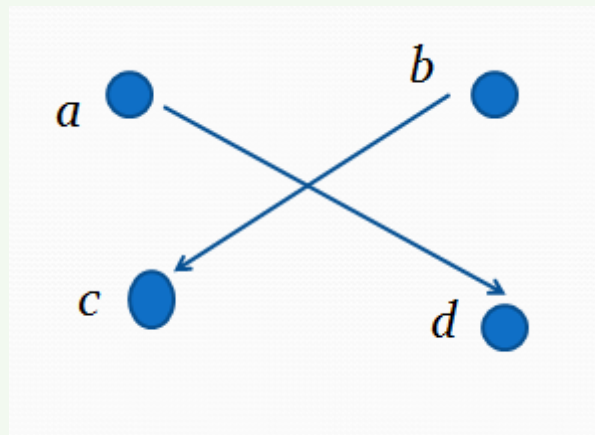
- **Reflexive:** a loop must be present at every vertex.
- **Symmetric:** if (x,y) is an edge then so is (y,x) .
- **Antisymmetric:** if (x,y) with $x \neq y$ is an edge, then (y,x) is not an edge.
- **Transitive:** if (x,y) and (y,z) are edges, then so is (x,z) .

Worked examples - determine properties of each digraph:

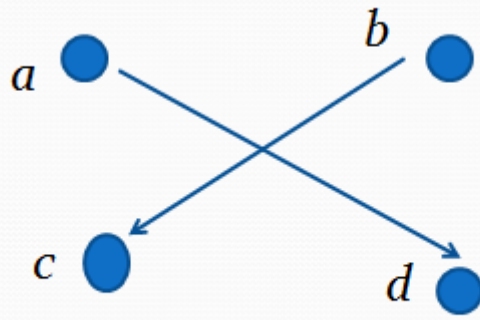
Example A: Reflexive? No — not every vertex has a loop. Symmetric? Yes (trivially). Antisymmetric? Yes (trivially). Transitive? Yes (trivially).



Example B: Reflexive? No — no loops. Symmetric? No. Antisymmetric? No. Transitive? No.



Example C: Reflexive? No. Symmetric? No. Antisymmetric? Yes. Transitive? Yes (trivially).



Example D: Reflexive? No. Symmetric? No. Antisymmetric? Yes. Transitive? Yes (trivially).

8. Equivalence Relations

8.1 Equivalence Relation and Equivalence Classes

Definition 1: A relation on a set A is called an **equivalence relation** if it is (i) reflexive, (ii) symmetric, and (iii) transitive.

Definition 2: Two elements a and b that are related by an equivalence relation are called **equivalent**. The notation $a \sim b$ is often used to denote that a and b are equivalent with respect to a particular equivalence relation.

Example: Congruence modulo m . Let m be an integer with $m > 1$. Show that $R = \{(a,b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers.

Recall that $a \equiv b \pmod{m}$ iff m divides $a - b$.

Reflexivity: $a \equiv a \pmod{m}$ since $a - a = 0$ is divisible by m ($0 = 0 \cdot m$).

Symmetry: suppose $a \equiv b \pmod{m}$, so $a - b = km$. Then $b - a = (-k)m$, so $b \equiv a \pmod{m}$.

Transitivity: suppose $a \equiv b$ and $b \equiv c \pmod{m}$. Then $a - b = km$ and $b - c = lm$. Adding: $a - c = (k + l)m$, so $a \equiv c \pmod{m}$.

Hence congruence modulo m is an equivalence relation. $\therefore$ proved.

Example: The “divides” relation is not an equivalence relation. On the set of positive integers, reflexivity and transitivity hold, but the relation is **not symmetric** (e.g. $2 \mid 4$ but $4 \nmid 2$). Hence “divides” is not an equivalence relation.

Reflexivity: $a \mid a$ for all a .

Transitivity: if $a \mid b$ and $b \mid c$ then $b = ak$ and $c = bl$, so $c = a(kl) \Rightarrow a \mid c$.

8.2 Congruence Modulo m (Equivalence Classes)

For a fixed modulus m , the equivalence class of an integer a under congruence mod m is the set of all integers congruent to a modulo m . This partitions the integers into exactly m disjoint classes (residue classes $0, 1, \dots, m-1$).

Example (mod 3): The classes are $[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}$, $[1] = \{\dots, -5, -2, 1, 4, 7, \dots\}$, $[2] = \{\dots, -4, -1, 2, 5, 8, \dots\}$. Every integer belongs to exactly one class.

8.3 Partitions

Partition: A **partition** of a set A is a collection of non-empty, pairwise-disjoint subsets whose union is A . The equivalence classes of an equivalence relation on A always form a partition of A , and conversely every partition gives rise to an equivalence relation (“two elements are related if they lie in the same block”).

Thus **equivalence relations and partitions are two views of the same idea**: an equivalence relation on A produces a partition of A , and each partition of A defines an equivalence relation on A .

9. Partial Orderings, POSET and Hasse Diagrams

9.1 Partial Ordering and POSET

Definition 1: A relation R on a set S is called a **partial ordering** (or partial order) if it is (i) reflexive, (ii) antisymmetric, and (iii) transitive. A set together with a partial ordering R is called a **partially ordered set**, or **poset**, denoted (S, R) . The members of S are called elements of the poset.

Example 1: $(\geq)$ is a partial ordering on the integers. Reflexivity: $a \geq a$.
Antisymmetry: if $a \geq b$ and $b \geq a$ then $a = b$. Transitivity: if $a \geq b$ and $b \geq c$ then $a \geq c$.

Example 2: The divisibility relation $(|)$ is a partial ordering on the positive integers.

Reflexivity: $a | a$ for all integers a .

Antisymmetry: if $a | b$ and $b | a$ (positive integers), then $a = b$.

Transitivity: if $a | b$ and $b | c$ then $b = ak$, $c = bl$, so $c = a(kl) \Rightarrow a | c$.

Hence $(\mathbb{Z}^+, |)$ is a poset.

Example 3: The inclusion relation $(\subseteq)$ is a partial ordering on the power set of a set S . Reflexivity: $A \subseteq A$. Antisymmetry: if $A \subseteq B$ and $B \subseteq A$ then $A = B$. Transitivity: if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

9.2 Comparability, Total Order and Well-Ordered Sets

Definition 2 (Comparability): The elements a and b of a poset $(S, \leq)$ are **comparable** if either $a \leq b$ or $b \leq a$. When neither holds, a and b are **incomparable**. (The symbol $\leq$ denotes the relation in any poset.)

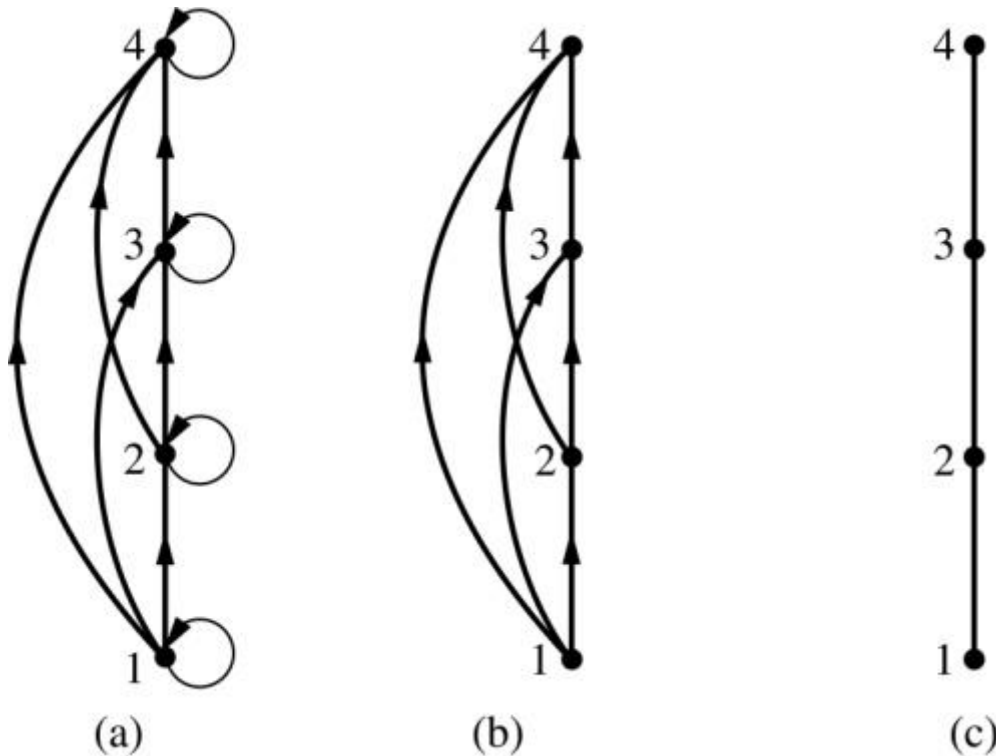
Definition 3 (Total Order): If $(S, \leq)$ is a poset and every two elements of S are comparable, S is called a **totally ordered** (or linearly ordered) set, and $\leq$ is called a **total order** (linear order). A totally ordered set is also called a **chain**.

Definition 4 (Well-Ordered): $(S, \leq)$ is a **well-ordered set** if it is a poset such that $\leq$ is a total ordering and every non-empty subset of S has a least element.

9.3 Hasse Diagrams

Definition: A **Hasse diagram** is a visual representation of a partial ordering that leaves out edges that must be present because of the reflexive and transitive properties.

In the figure below, part (a) shows the full directed graph of a partial ordering; the loops due to the reflexive property are deleted in (b); the edges that must be present due to the transitive property are deleted in (c). The Hasse diagram is (c).



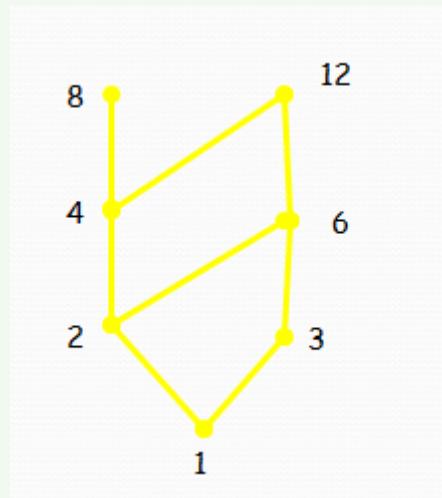
Construction of a Hasse diagram: (a) full digraph, (b) loops removed, (c) transitive edges removed — the Hasse diagram.

Procedure for Constructing a Hasse Diagram: To represent a finite poset $(S, \leq)$ using a Hasse diagram, start with the directed graph of the relation:

1. Remove the loops (a, a) present at every vertex due to the reflexive property.
2. Remove all edges (x, y) for which there is an element $z \in S$ such that $x \leq z$ and $z \leq y$ (edges that must be present due to transitivity).
3. Arrange each edge so that its initial vertex is below the terminal vertex, and remove all arrows (all edges point upwards).

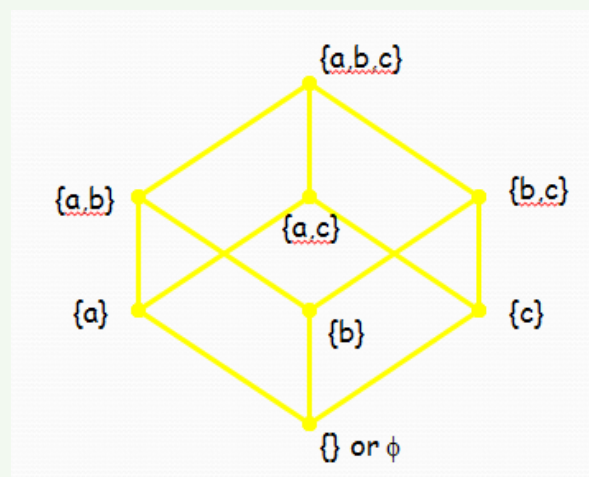
Example 1: Hasse diagram of the poset $(\{1, 2, 3, 4\}, \leq)$ — a single vertical chain $1 < 2 < 3 < 4$.

Example 2: Draw the Hasse diagram representing the partial ordering $\{(a,b) \mid a \text{ divides } b\}$ on $\{1,2,3,4,6,8,12\}$.



Hasse diagram for the divisibility poset on $\{1,2,3,4,6,8,12\}$.

Example 3: Draw the Hasse diagram for the partial ordering $\{(A,B) \mid A \subseteq B\}$ on the power set $P(S)$, where $S = \{a,b,c\}$.

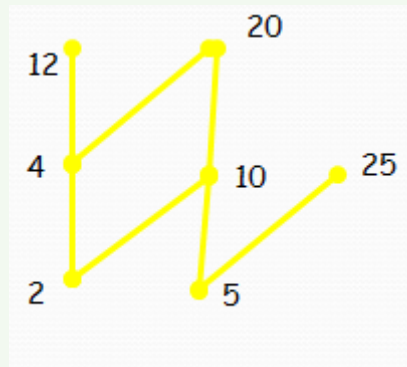


Hasse diagram of the power-set lattice $(P(\{a,b,c\}), \subseteq)$.

9.4 Minimal, Maximal, Least and Greatest Elements

Minimal and Maximal: In a poset S , an element a is **minimal** if no other element strictly precedes a ; an element b is **maximal** if no element strictly succeeds b . They are respectively the bottom and top elements of the diagram. A poset can have more than one minimal and more than one maximal element.

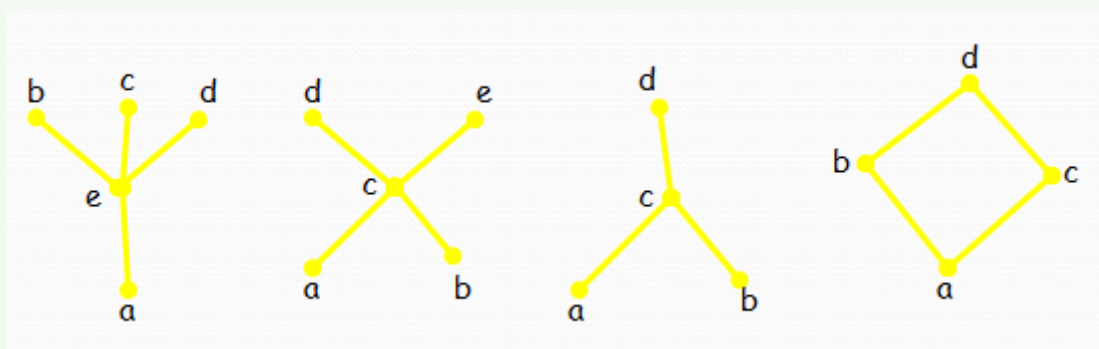
Example: In the poset $(\{2,4,5,10,12,20,25\}, |)$, the maximal elements are 12, 20 and 25, and the minimal elements are 2 and 5.



Maximal and minimal elements of the divisibility poset.

Least and Greatest: An element a in a poset S is the **least** (first) element if a precedes every other element of S ; an element b is the **greatest** (last) element if b succeeds every other element. When they exist, the greatest and least elements are **unique**.

Examples: In the four Hasse diagrams below: (i) least element is a , no greatest element; (ii) no least element and no greatest element; (iii) no least element and d is the greatest element; (iv) least element is a and greatest element is d .



Four posets illustrating greatest and least elements.

9.5 Upper and Lower Bounds, Supremum and Infimum

Upper and Lower Bounds: Let A be a subset of a poset S . An element u in S is an **upper bound** of A if u succeeds every element of A (u relates to every element of A). An element l in S is a **lower bound** of A if l precedes every element of A .

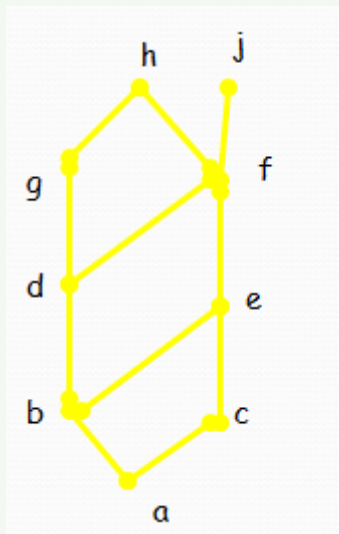
Supremum: If an upper bound of A precedes every other upper bound of A , it is called the **supremum** or **least upper bound** of A , denoted $\sup(A)$ or $\text{lub}(A)$.

Infimum: If a lower bound of A succeeds every other lower bound of A , it is called the **infimum** or **greatest lower bound** of A , denoted $\inf(A)$ or $\text{glb}(A)$.

Example (from slides): In a given poset, for the subset $\{3,4\}$: 4, 5 and 6 are upper bounds ($3R4, 4R4 \Rightarrow 4$; $3R5, 4R5 \Rightarrow 5$; $3R6, 4R6 \Rightarrow 6$), and 1, 2, 3 are lower bounds.

For the subsets $\{1,3\}$ and $\{4,5\}$ of another poset: lower bounds of $\{1,3\} = \{3,4,5\}$; lower bounds of $\{4,5\} = \{5\}$; upper bound of $\{1,3\} = \{1\}$; upper bounds of $\{4,5\} = \{1,2,3,4\}$.

Example (Hasse diagram): Find the lower and upper bounds of the subsets $\{a,b,c\}$, $\{j,h\}$ and $\{a,c,d,f\}$ in the poset with the given Hasse diagram.



Poset with elements a through j for bound computations.

10. Closures of Relations and Warshall's Algorithm

10.1 Closure of a Relation

Definition: Let R be a relation on a set A that may or may not have some property P (e.g. reflexive, symmetric, transitive). If there is a relation S with property P containing R such that S is a subset of every relation with property P containing R , then S is called the **closure of R with respect to P** .

Note: The closure may or may not exist. The closure is the *smallest* relation with property P that contains R .

10.2 Reflexive and Symmetric Closure

Theorem (Reflexive Closure): Let R be a relation on A . The reflexive closure of R , denoted $r(R)$, is $R \cup \Delta$ where $\Delta = \{(a,a) : a \in A\}$. Equivalently, add loops to all vertices in the digraph, or put 1's on the diagonal of the connection matrix of R .

Example 1: What is the reflexive closure of $R = \{(a,b) \mid a < b\}$ on the integers? (Answer: add all pairs (a,a) , giving $\leq$.)

Inverse of a Relation: The inverse of R is $R^{-1} = \{ \langle y, x \rangle \mid \langle x, y \rangle \in R \}$. To get R^{-1} , reverse all arcs in the digraph, or take the transpose M^T of the connection matrix M . This relation is sometimes denoted R^T or R^c and called the converse of R .

Theorem (Symmetric Closure): Let R be a relation on A . The symmetric closure of R , denoted $s(R)$, is the relation $R \cup R^{-1}$.

Example 2: What is the symmetric closure of $R = \{(a,b) \mid a > b\}$ on the positive integers? (Answer: $R \cup R^{-1} = \{(a,b) : a \neq b\}$.)

Worked example: Let $R = \{(1,2), (2,3), (3,1)\}$ on $A = \{1,2,3\}$. Find the reflexive and symmetric closures.

Reflexive closure: $S = R \cup \Delta = \{(1,2), (2,3), (3,1)\} \cup \{(1,1), (2,2), (3,3)\}$
 $= \{(1,1), (1,2), (2,2), (2,3), (3,1), (3,3)\}$.

Symmetric closure: $S = R \cup R^{-1} = \{(1,2), (2,3), (3,1)\} \cup \{(2,1), (3,2), (1,3)\}$
 $= \{(1,2), (2,1), (2,3), (3,2), (3,1), (1,3)\}$.

10.3 Transitive Closure

Definition: The transitive closure of a relation R , $t(R)$, is the smallest transitive relation containing R . Recall: R is transitive iff $R^n \subseteq R$ for all n . Hence if there is a path from x to y then there must be an arc from x to y , i.e. $\langle x, y \rangle \in t(R)$.

Theorem: Let M_R be the zero-one matrix of the relation R on a set with n elements. Then the zero-one matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \dots \vee M_R^{[n]}$$

Worked example: Let $R = \{(1,2), (2,3), (3,1)\}$ on $A = \{1,2,3\}$. Find the transitive closure.

(i) **By composition of relations:**

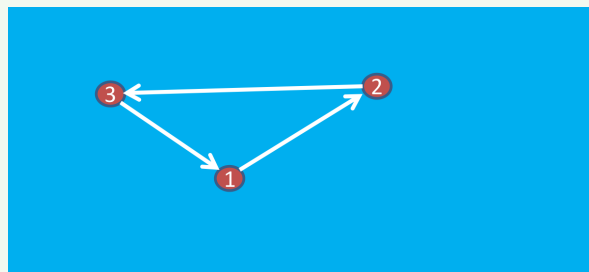
$$R^2 = \{(1,3), (2,1), (3,2)\}, \quad R^3 = \{(1,1), (2,2), (3,3)\}.$$

$$t(R) = R \cup R^2 \cup R^3 = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}.$$

(ii) **By matrices:** $S = M \vee M^{[2]} \vee M^{[3]}$

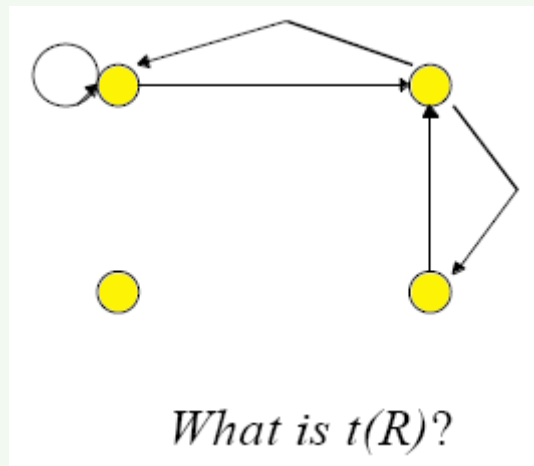
$$\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \vee \begin{array}{ccc} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \vee \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} = \begin{array}{ccc} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array}$$

which corresponds to all nine ordered pairs. (iii) **By digraph:** the closure adds a loop at every vertex and an arc in both directions between every pair.



The digraph of $R = \{(1,2), (2,3), (3,1)\}$: a directed 3-cycle.

Example (digraph): What is $t(R)$ for the relation shown below?



A digraph whose transitive closure is required.

10.4 Warshall's Algorithm

Warshall's algorithm computes the transitive closure efficiently. Let R be a relation with zero-one matrix $W^0 = M_R$. Define $W^k = [w_{ij}^k]$ where

$$w_{ij}^k = w_{ij}^{k-1} \text{ OR } (w_{ik}^{k-1} \text{ AND } w_{kj}^{k-1})$$

That is, $W^k[i][j] = 1$ if $W^{k-1}[i][j]$ is 1, or both $W^{k-1}[i][k]$ and $W^{k-1}[k][j]$ are 1. After n steps, W^n is the matrix of the transitive closure.

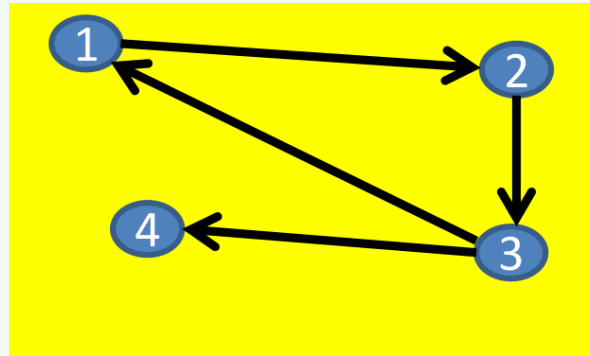
Example 1: Use Warshall's algorithm to find the transitive closure of R on $A = \{1,2,3\}$ where $R = \{(1,1),(1,3),(2,2),(3,1),(3,2)\}$.

$$W^0 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

W^0		W^1		W^2		W^3
1 0 1	$\rightarrow (k=1)$	1 0 1	$\rightarrow (k=2)$	1 0 1	$\rightarrow (k=3)$	1 1 1
0 1 0		0 1 0		0 1 0		0 1 0
1 1 0		1 1 1		1 1 1		1 1 1

Thus the transitive closure is $\{(1,1),(1,2),(1,3),(2,2),(3,1),(3,2),(3,3)\}$.

Example 2: Use Warshall's algorithm to find the transitive closure of R on $A = \{1,2,3,4\}$ where the digraph is as shown:



Digraph of R with edges $1 \rightarrow 2$, $1 \rightarrow 3$, $3 \rightarrow 4$, $3 \rightarrow 1$.

$$W^0 = \begin{bmatrix} [0,1,0,0], & [0,0,1,0], & [1,0,0,1], & [0,0,0,0] \end{bmatrix}$$

W^1	W^2	W^3	W^4
0 1 0 0	0 1 1 0	1 1 1 1	1 1 1 1
0 0 1 0	0 0 1 0	1 1 1 1	1 1 1 1
1 1 0 1	1 1 1 1	1 1 1 1	1 1 1 1
0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Since $W^4 = W^3$, the transitive closure is the relation whose matrix is W^3 .

11. Lattices

11.1 Lattice - Definition, Meet and Join

Definition: A partially ordered set in which **every pair of elements** has both a **least upper bound** and a **greatest lower bound** is called a **lattice**.

- The least upper bound is called the **join**, denoted $a \vee b$ (or $a \vee b$).
- The greatest lower bound is called the **meet**, denoted $a \wedge b$ (or $a \wedge b$).

Example 1: Is the poset $(\mathbb{Z}^+, |)$ a lattice? Let a and b be two positive integers. The least upper bound of a and b is their **L.C.M.** and the greatest lower bound is their **G.C.D.** Since both always exist, this poset **is a lattice**.

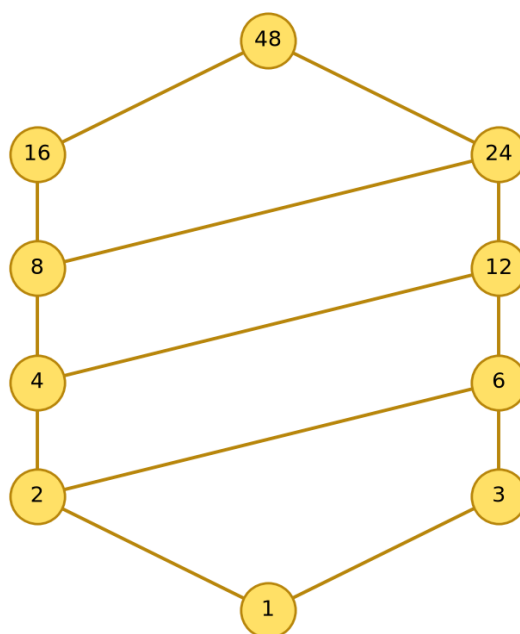
Example 2: Determine whether $(\{1,2,3,4,5\}, |)$ and $(\{1,2,4,8,16\}, |)$ are lattices. (The first is **not** a lattice — e.g. 2 and 3 have no upper bound within the set; the second **is** a lattice — a chain.)

Example 3: Is $(P(S), \subseteq)$ a lattice? Yes — for any two subsets A, B : $\text{lub} = A \cup B$ and $\text{glb} = A \cap B$.

Theorem: For any $a, b \in L$ where L is a lattice: $a \leq a \vee b$ and $a \wedge b \leq a$.

Examples of divisor lattices (under the divisibility relation):

Hasse diagram of divisors of 48



The divisor lattice $(D(48), |)$ — every pair has a meet (gcd) and join (lcm).

11.2 Lattice Axioms and Duality

Algebraic definition: Let L be a non-empty set closed under two binary operations, meet ($\wedge$) and join ($\vee$). Then L is a lattice if it satisfies:

- **L1 - Commutative Law:** $a \wedge b = b \wedge a$ and $a \vee b = b \vee a$.
- **L2 - Associative Law:** $(a \wedge b) \wedge c = a \wedge (b \wedge c)$ and $(a \vee b) \vee c = a \vee (b \vee c)$.
- **L3 - Absorption Law:** $a \wedge (a \vee b) = a$ and $a \vee (a \wedge b) = a$.

where a, b, c are elements in L .

Proof - meet and join are commutative:

$$a \wedge b = \text{glb}\{a, b\} = \text{glb}\{b, a\} = b \wedge a.$$

$$a \vee b = \text{lub}\{a, b\} = \text{lub}\{b, a\} = b \vee a.$$

Proof - idempotent:

$$a \wedge a = \text{glb}\{a, a\} = \text{glb}\{a\} = a. \quad a \vee a = \text{lub}\{a, a\} = \text{lub}\{a\} = a.$$

Proof - associative (join): Since $(a \vee b) \vee c = \text{lub}\{a \vee b, c\}$, we have $a \vee b \leq (a \vee b) \vee c$ and $c \leq (a \vee b) \vee c$. Since $a \leq a \vee b$ and $b \leq a \vee b$, by transitivity a, b, c are all $\leq (a \vee b) \vee c$, so $a \vee (b \vee c) \leq (a \vee b) \vee c$. Symmetrically, $(a \vee b) \vee c \leq a \vee (b \vee c)$. By antisymmetry, $(a \vee b) \vee c = a \vee (b \vee c)$.

Proof - absorption: Since $a \wedge b \leq a$ and $a \leq a$, we get $a \vee (a \wedge b) \leq a$; also $a \leq a \vee (a \wedge b)$ by definition of lub. Hence $a \vee (a \wedge b) = a$.

Duality: The **dual** of any statement in a lattice is the statement obtained by interchanging meet and join. Example: the dual of $a \wedge (b \vee a) = a \vee a$ is $a \vee (b \wedge a) = a \wedge a$.

Theorem: For any lattice and $a, b \in L$, if $a \leq b$ and $c \leq d$ then $a \vee c \leq b \vee d$ and $a \wedge c \leq b \wedge d$.

11.3 Sublattices and Distributive Lattices

Sublattice: Let S be a non-empty subset of a lattice L . Then S is a **sublattice** of L if S itself is a lattice (closed under the meet and join of L).

Example 1: In $(P(S), \subseteq)$, only those collections of $P(S)$ that are closed under both $\cup$ and $\cap$ are sublattices. If $S = \{a, b, c, d\}$, then $\{\emptyset, \{a\}, \{a, c\}, \{c\}, \{a, b, c\}\}$ is a sublattice, but $\{\emptyset, \{a\}, \{b\}, \{c\}, \{b, c\}\}$ is **not**, because $\{a\} \cup \{c\} = \{a, c\}$ is missing.

Example 2: If $L = D(30)$, then $D(6)$, $D(10)$, $D(15)$ are sublattices. In general, if $m \mid n$ then $D(m)$ is a sublattice of $D(n)$.

Distributive Lattice: A lattice L is **distributive** if for all $a, b, c \in L$:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

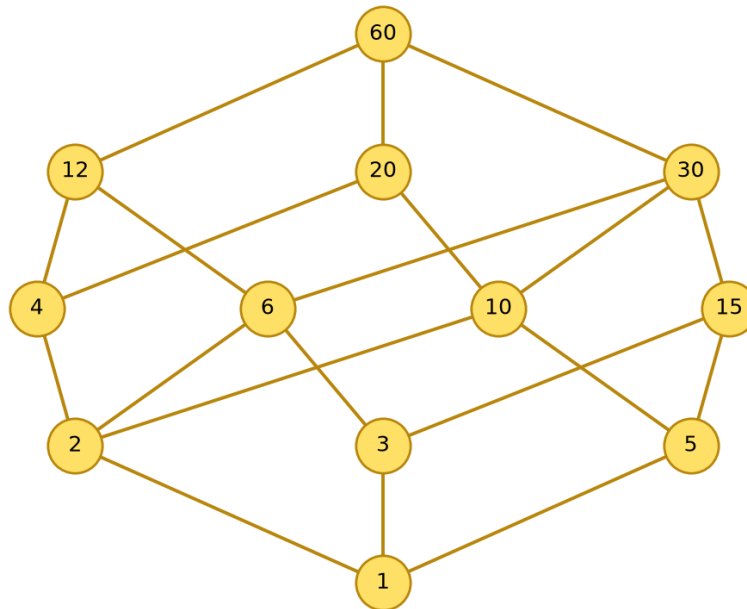
$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Otherwise L is called non-distributive.

Theorem: In any distributive lattice, $a \wedge x = a \wedge y$ and $a \vee x = a \vee y$ together imply $x = y$.

$$x = x \vee (x \wedge a) = x \vee (y \wedge a) = (x \vee y) \wedge (x \vee a) = (y \vee x) \wedge (y \vee a) = y \vee (x \wedge a) = y \vee (y \wedge a) = y.$$

Hasse diagram of divisors of 60



The divisor lattice $(D(60), |)$ is a distributive lattice.

11.4 Bounded and Complemented Lattices

Bounded Lattice: A lattice L is **bounded** if it has both a lower bound, denoted 0 , such that $0 \leq x$ for all $x \in L$, and an upper bound, denoted 1 , such that $x \leq 1$ for all $x \in L$. The bounds 0 and 1 are called **universal bounds**. In a bounded lattice: $a \vee 1 = 1$, $a \wedge 1 = a$, $a \vee 0 = a$, $a \wedge 0 = 0$.

Theorem: Every finite lattice L is bounded. Proof: if $L = \{a_1, a_2, \dots, a_n\}$ then $a_1 \wedge \dots \wedge a_n$ and $a_1 \vee \dots \vee a_n$ are lower and upper bounds of L .

Examples: $L = \{1, 2, 3, 6\}$ with $(L, |)$ is a bounded lattice (lower bound 1, upper bound 6). The lattice $(\mathbb{N}, \leq)$ is **not** bounded because it has no upper bound (though 1 is a lower bound).

Complement: Let L be a bounded lattice with lower bound 0 and upper bound 1, and let $a \in L$. An element $x \in L$ is a **complement** of a if $a \vee x = 1$ and $a \wedge x = 0$. Complement of 0 = 1; complement of 1 = 0. Complements need not exist and need not be unique.

Complemented Lattice: A lattice L is **complemented** if it is bounded and every element in L has a complement.

Example: The lattice $(\mathcal{P}(S), \subseteq)$ of the power set of any set S is complemented; in this lattice each element has a unique complement (its set complement).

Theorem: In a distributive lattice, if an element has a complement, then this complement is unique. (Proof uses $a \vee x = 1$, $a \wedge x = 0$ for two supposed complements and the distributive law to show they are equal.)

11.5 Complete and Modular Lattices

Complete Lattice: A lattice L is **complete** if every subset of L has a glb and a lub.

- A complete lattice is always bounded, but a bounded lattice need not be complete.
- Every finite lattice is complete.

Example: The lattice $L = \{0, 1, 2, 3, 6\}$ under the divisibility relation is a complete lattice.

Modular Lattice: A lattice L is **modular** if $a \leq c$ implies $a \vee (b \wedge c) = (a \vee b) \wedge c$.

Example: Every distributive lattice is modular. (If $a \leq c$ then $a \vee c = c$, and $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c) = (a \vee b) \wedge c$.)

Theorem (De Morgan in a lattice): If $(L, \vee, \wedge)$ is a distributive complemented lattice, then De Morgan's laws hold:

$$(a \vee b)' = a' \wedge b' \quad \text{and} \quad (a \wedge b)' = a' \vee b'$$

To prove $(a \vee b)' = a' \wedge b'$, show that $a' \wedge b'$ is the complement of $a \vee b$: $(a \vee b) \vee (a' \wedge b') = 1$ and $(a \vee b) \wedge (a' \wedge b') = 0$.

11.6 Homomorphism and Isomorphism

Homomorphism: Let L and M be lattices. A mapping $f : L \rightarrow M$ is a **homomorphism** if it satisfies $f(x \vee y) = f(x) \vee f(y)$ and $f(x \wedge y) = f(x) \wedge f(y)$.

Isomorphism: Let L and M be lattices. A mapping $f : L \rightarrow M$ is an **isomorphism** if it satisfies (i) f is one-to-one, (ii) f is onto, and (iii) f is a homomorphism.

12. Question Bank - Unit I (Solved)

Q1. Use Warshall's algorithm to calculate the transitive closure of relation R on A = {1,2,3} where R = {(1,1),(1,3),(2,2),(3,1),(3,2)}.

Solution. The initial matrix is

$$W^0 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

Applying Warshall's algorithm:

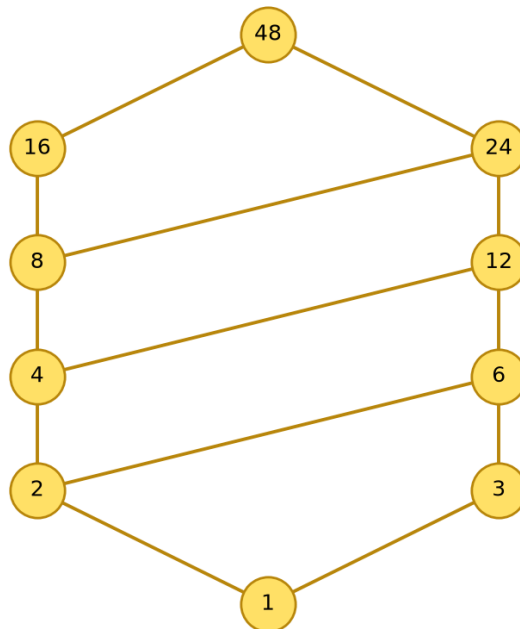
$$\begin{array}{ccc} W^1 = & W^2 = & W^3 = \\ \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \end{array}$$

Hence the transitive closure is $R^* = \{(1,1),(1,2),(1,3),(2,2),(3,1),(3,2),(3,3)\}$.

Q2. Define poset. Draw the Hasse diagram of all divisors of 48. Is it a lattice?

Solution. A **poset** is a set S together with a relation R that is reflexive, antisymmetric and transitive, written (S, R). The divisors of 48 are 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

Hasse diagram of divisors of 48



Hasse diagram of the divisors of 48 under the “divides” relation.

Is it a lattice? Yes. For any two divisors a, b of 48, their gcd (greatest lower bound) and lcm (least upper bound) are also divisors of 48, so every pair has a meet and a join. Hence $(D(48), |)$ is a lattice.

Q3. Prove using a truth table that $(P \vee Q) \wedge (\neg P \vee R)$ is logically equivalent to $(P \wedge R) \vee (Q \wedge \neg R)$.

Solution. Construct the truth table:

P	Q	R	$P \vee Q$	$\neg P \vee R$	$(P \vee Q) \wedge (\neg P \vee R)$	$P \wedge R$	$Q \wedge \neg R$	$(P \wedge R) \vee (Q \wedge \neg R)$
T	T	T	T	T	T	T	F	T
T	T	F	T	F	F	F	T	T
T	F	T	T	T	T	T	F	T
T	F	F	T	F	F	F	F	F
F	T	T	T	T	T	F	F	F
F	T	F	T	T	T	F	T	T
F	F	T	F	T	F	F	F	F
F	F	F	F	T	F	F	F	F

Observation. Comparing the two compound columns, they agree in 5 of the 8 rows but **differ** in rows (T,T,F), (F,T,T) and (F,T,F). Therefore, exactly as printed, the two propositions are **not** logically equivalent — the term on the right should be **$(Q \wedge \neg P)$** rather than $(Q \wedge \neg R)$. With that correction the two columns match in every row, and we obtain the standard equivalence:

$$(P \vee Q) \wedge (\neg P \vee R) \equiv (P \wedge R) \vee (\neg P \wedge Q)$$

Q4. Apply Warshall's algorithm to compute the transitive closure of a relation represented by the adjacency matrix

```

0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0

```

Solution.

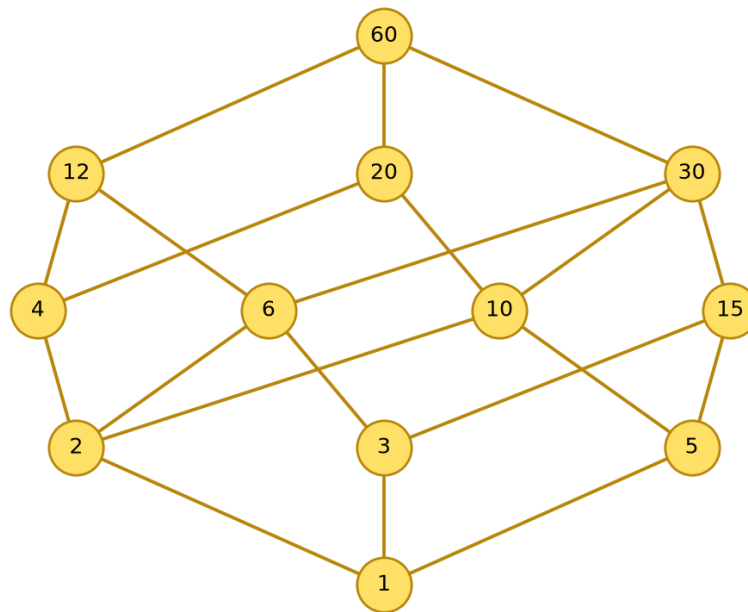
$$\begin{array}{cccc}
 W^0 = & W^1 = & W^2 = & W^3 = W^4 = \\
 0\ 1\ 0\ 1 & 0\ 1\ 0\ 1 & 1\ 1\ 1\ 1 & 1\ 1\ 1\ 1 \\
 1\ 0\ 1\ 0 & 1\ 1\ 1\ 1 & 1\ 1\ 1\ 1 & 1\ 1\ 1\ 1 \\
 0\ 1\ 0\ 1 & 0\ 1\ 0\ 1 & 1\ 1\ 1\ 1 & 1\ 1\ 1\ 1 \\
 1\ 0\ 1\ 0 & 1\ 1\ 1\ 1 & 1\ 1\ 1\ 1 & 1\ 1\ 1\ 1
 \end{array}$$

The transitive closure is the **universal relation**: every element is related to every element.

Q5. Define a poset. Draw the Hasse diagram for the divisors of 60 under the “divides” relation and verify whether that forms a poset.

Solution. A poset (S, R) is a set with a reflexive, antisymmetric and transitive relation. The divisors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60.

Hasse diagram of divisors of 60

*Hasse diagram of the divisors of 60 under the “divides” relation.***Verification:**

- **Reflexive:** $a \mid a$ for every divisor a .
- **Antisymmetric:** if $a \mid b$ and $b \mid a$ then $a = b$.
- **Transitive:** if $a \mid b$ and $b \mid c$ then $a \mid c$.

All three properties hold, so $(D(60), \mid)$ is a poset.

Q6. Prove by mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n .

Solution.

Basis ($n = 1$): $6^3 + 7^3 = 216 + 343 = 559 = 43 \times 13$. Divisible by 43. ✓

Induction hypothesis: assume $6^{k+2} + 7^{2k+1} = 43m$ for some integer m .

Inductive step:

$$\begin{aligned}
 6^{k+3} + 7^{2k+3} &= 6 \cdot 6^{k+2} + 49 \cdot 7^{2k+1} \\
 &= 6 \cdot 6^{k+2} + 6 \cdot 7^{2k+1} + 43 \cdot 7^{2k+1} \\
 &= 6(6^{k+2} + 7^{2k+1}) + 43 \cdot 7^{2k+1} \\
 &= 6(43m) + 43 \cdot 7^{2k+1} = 43(6m + 7^{2k+1}).
 \end{aligned}$$

This is divisible by 43. Hence by the principle of mathematical induction, $6^{n+2} + 7^{2n+1}$ is divisible by 43 for all positive integers n . □

Q7. Using Warshall's algorithm, find the transitive closure of the relation $R = \{(1,4), (2,1), (2,3), (3,1), (3,4), (4,3)\}$ on $A = \{1,2,3,4\}$.

Solution.

$$\begin{array}{cccc}
 W^0 = & W^1 = & W^3 = & W^4 = \\
 0001 & 0001 & 0001 & 1011 \\
 1010 & 1011 & 1011 & 1011 \\
 1001 & 1001 & 1001 & 1011 \\
 0010 & 0010 & 1011 & 1011
 \end{array}
 \quad W^2 = W^1$$

Hence the transitive closure is $R^* = \{(1,1), (1,3), (1,4), (2,1), (2,3), (2,4), (3,1), (3,3), (3,4), (4,1), (4,3), (4,4)\}$.

Q8. Show that the division relation on the set of integers is a partial ordering relation.

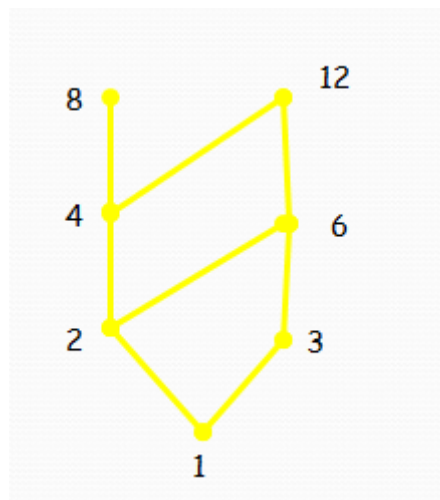
Solution. On the set of **positive** integers the divisibility relation (" $|$ ") is a partial ordering:

- **Reflexive:** $a | a$ for every integer a .
- **Antisymmetric:** if $a | b$ and $b | a$, then $a = b$.
- **Transitive:** if $a | b$ and $b | c$, then $b = ak$ and $c = bl$, so $c = a(kl)$, i.e. $a | c$.

Therefore $(\mathbb{Z}^+, |)$ is a poset. **Caveat:** on *all* integers (including negatives) antisymmetry fails, since e.g. $2 | -2$ and $-2 | 2$ but $2 \neq -2$. Hence divisibility is a partial ordering only on the positive integers (or on non-negative integers).

Q9. Draw the Hasse diagram representing the partial ordering $\{(a,b) : a \text{ divides } b\}$ on $\{1,2,3,4,6,8,12\}$.

Solution.



Hasse diagram of the divisibility poset on $\{1,2,3,4,6,8,12\}$.

13. Quick Revision Summary

Logic

- A **proposition** has a definite truth value T or F; questions, commands, exclamations and open sentences are not propositions.
- **Connectives:** $\neg$ (not), $\wedge$ (and), $\vee$ (or, inclusive), $\oplus$ (exclusive or), $\rightarrow$ (if...then), $\leftrightarrow$ (iff).
- **$p \rightarrow q$ is false only when p is true and q is false.**
- **Contrapositive** $\neg q \rightarrow \neg p$ is equivalent to $p \rightarrow q$; the **converse** $q \rightarrow p$ is not.
- **Tautology** always true; **contradiction** always false; **contingency** neither.
- **De Morgan:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Predicate Logic

- A **predicate** $P(x)$ becomes a proposition when x is assigned a value; the **domain** matters.
- $\forall x P(x)$ — true for every element; $\exists x P(x)$ — true for at least one.
- Negation: $\neg \forall x P(x) \equiv \exists x \neg P(x)$; $\neg \exists x P(x) \equiv \forall x \neg P(x)$.
- Order of nested quantifiers matters: $\forall x \exists y P(x,y)$ is not the same as $\exists y \forall x P(x,y)$.

Rules of Inference

- Modus Ponens ($p, p \rightarrow q \Rightarrow q$); Modus Tollens ($\neg q, p \rightarrow q \Rightarrow \neg p$); Hypothetical Syllogism ($p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$); Disjunctive Syllogism ($p \vee q, \neg p \Rightarrow q$); Addition; Simplification; Conjunction; Resolution.
- Fallacies: affirming the consequent and denying the antecedent are **invalid**.
- Quantified rules: UI, UG, EI, EG.

Proof Techniques

- Direct, Contrapositive ($\neg q \rightarrow \neg p$), Contradiction, Cases, Biconditional (prove both directions), Existence, Uniqueness.
- **Induction:** Basis $P(1) \rightarrow$ Hypothesis $P(k) \rightarrow$ Inductive Step $P(k+1)$. **Strong induction** assumes $P(1) \dots P(k)$.

Relations

- **Reflexive** ($a a a$), **Symmetric** ($a R b \Rightarrow b R a$), **Transitive** ($a R b, b R c \Rightarrow a R c$), **Antisymmetric** ($a R b, b R a \Rightarrow a=b$).
- **Equivalence** = reflexive + symmetric + transitive; **Partial order** = reflexive + antisymmetric + transitive.
- Representation: zero-one **matrix** M_R and **digraph**. Reflexive $\rightarrow$ diagonal all 1; symmetric $\rightarrow M = M^T$.
- Equivalence classes partition the set; congruence mod m is the standard example.

POSET, Hasse, Closure, Lattice

- **Hasse diagram:** remove loops and transitive edges, draw edges upward without arrows.

- **Minimal/maximal** (no element strictly before/after); **least/greatest** (unique when they exist); **upper/lower bounds, supremum (lub)** and **infimum (glb)**.
- **Closures:** reflexive $r(R) = R \cup \Delta$; symmetric $s(R) = R \cup R^{-1}$; transitive $t(R) = R \cup R^2 \cup \dots \cup R^n$.
- **Warshall's algorithm:** $W^k[i][j] = W^{k-1}[i][j] \text{ OR } (W^{k-1}[i][k] \text{ AND } W^{k-1}[k][j])$; W^n gives the transitive closure.
- **Lattice:** every pair has a join (lub) and a meet (glb); commutative, associative, absorption laws; distributive, bounded (0, 1), complemented, complete, modular types; homomorphism and isomorphism.

End of Unit I - Logic and Relations. Complete notes compiled from the lecture notes, slides and question bank of Discrete Mathematics and Graph Theory (DMGT).